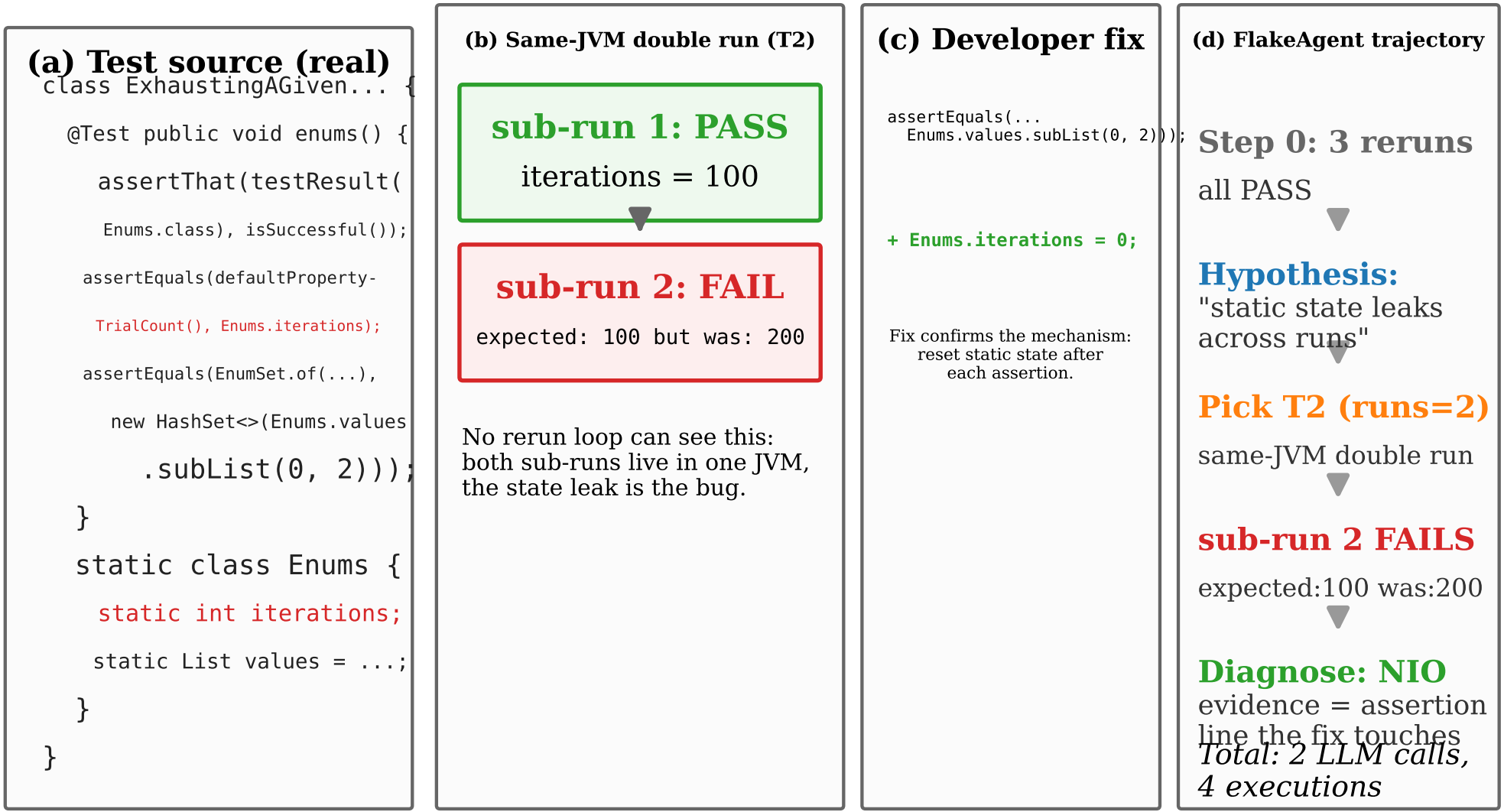

(a) Test source (real)
class ExhaustingAGiven... {
@Test public void enums() {
assertThat(testResult(
Enums.class), isSuccessful());
assertEquals(defaultProperty-
TrialCount(), Enums.iterations);
assertEquals(EnumSet.of(...),
new HashSet<>(Enums.values
.subList(0, 2)));
}
static class Enums {
static int iterations;
static List values = ...;
}
}
(b) Same-JVM double run (T2)
sub-run 1: PASS
iterations = 100
sub-run 2: FAIL
expected: 100 but was: 200
No rerun loop can see this:
both sub-runs live in one JVM,
the state leak is the bug.
(c) Developer fix
assertEquals(...
Enums.values.subList(0, 2)));
+ Enums.iterations = 0;
Fix confirms the mechanism:
reset static state after
each assertion.
(d) FlakeAgent trajectory
Step 0: 3 reruns
all PASS
Hypothesis:
"static state leaks
across runs"
Pick T2 (runs=2)
same-JVM double run
sub-run 2 FAILS
expected:100 was:200
Diagnose: NIO
evidence = assertion
line the fix touches
Total: 2 LLM calls,
4 executions